

TO: Dr. Azade Farshad & Doctoral Selection Committee · Dept. of Computer Science, Aalto University / ELLIS Institute Finland
SUBJECT: Application for Doctoral Studies (Autumn 2026) — Bidirectional Foundation Architectures for Continuous Sign Language AI
APPLICATION DOSSIER & REPOSITORY LINKS:
► **Interactive 15-DOF Kinematic PoC (WebGL):** joeytribb.github.io/sign-language-kinematics/anatomical_finger.html
► **Full 14-Page Technical Proposal (Direct PDF):** joeytribb.github.io/sign-language-kinematics/Proposal.pdf · [GitHub Repo]

Dear Dr. Farshad and Members of the Selection Committee,

I am writing to apply for the Doctoral Programme in Science at Aalto University and the ELLIS Institute Finland under the supervision of **Dr. Azade Farshad**. My proposed research directly addresses ELLIS Finland's strategic mandate in *AI for Health & Assistive Digital Twins* by addressing a critical, unmet challenge in modern artificial intelligence: the persistent technological exclusion of over 70 million Deaf individuals worldwide who rely on sign languages as their native medium of thought and communication.

1 The Grand Challenge: Why Sign Language AI Remains Unsolved

Over 70 million Deaf individuals rely on sign languages, yet mainstream AI has struggled to deliver robust solutions. While foundation models have transformed text and audio processing, continuous sign language technology remains constrained between 2D video models that lack spatial grounding and rigid animated avatars that native signers find unnatural. The core challenge is representational: machine learning standardly treats sign as flat 1D text tokens or 2D pixel grids. In reality, sign languages are **multi-channel, spatial, and simultaneous**—coordinating 15-DOF finger dexterity, facial grammar, and 3D reference loci (*R*-loci) in physical space. A 3-centimeter positional shift alters the meaning from MOTHER to FATHER, and temporal context truncation collapses spatial agreement verbs entirely. My doctoral objective is to address this fundamental bottleneck: **developing a closed-loop, 3D spatial foundation architecture that unifies native grammatical precision with the biomechanical naturalism of human physical dynamics.**

2 Intellectual Genesis & Demonstrated Technical Foundations

My motivation to solve this challenge originates from continuous physical control and stochastic reinforcement learning. In my M.Sc. thesis (*Autonomous High-Frequency Trading via Deep Reinforcement Learning*, open-sourced as *bithax* on GitHub), I investigated how agents optimize sequential decision-making in continuous, non-stationary state spaces under non-linear constraints and stochastic noise. In research mentorship with Dr. Marco Marini (Sapienza University of Rome) analyzing sign language phonology and kinematics, I realized sign generation is not a computer vision task—it is a **continuous trajectory control problem operating on an articulated 3D kinematic skeleton, governed by a discrete spatial grammar**. Pure data-driven models fail because unconstrained networks lack physical priors, deforming joints into the “uncanny valley.” To evaluate how physical constraints prevent hallucinations, I developed an interactive 15-DOF WebGL kinematics engine (anatomical_finger.html). Modeling 3-phalanx kinematic chains per digit under strict anatomical joint limits (MCP flexion/abduction, PIP, DIP) alongside constrained Cyclic Coordinate Descent (CCD) IK, it compiles symbolic phonemes into deterministic 3D hand configurations in $\mathcal{O}(1)$ time, confirming that deterministic geometric priors reliably stabilize articulated motion.

3 Proposed Methodology & Scientific Alignment with Dr. Farshad

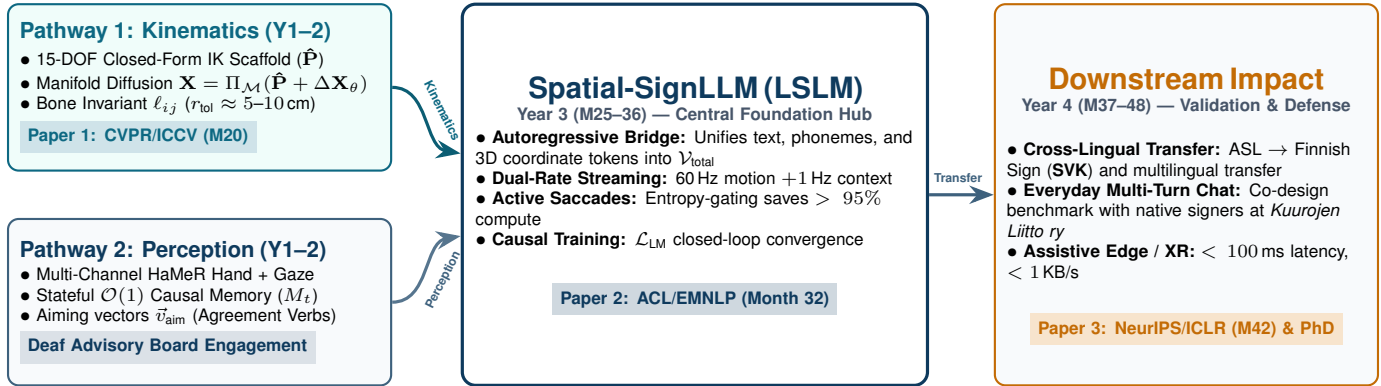
To unify grammatical precision with biological naturalism, I propose a closed-loop bidirectional foundation architecture:

- **Forward Pass (Text → 3D Avatar):** Translates discourse into a spatial scene graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with assigned reference loci (*R*-loci), compiles symbolic phonemes into deterministic 15-DOF procedural scaffolding ($\hat{\mathbf{P}}$) via closed-form IK, and applies **Manifold-Projected Residual Diffusion** $\mathbf{X} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta \mathbf{X}_{\theta}(\cdot \mid \mathcal{G}))$ subject to bone-length preservation $\|\mathbf{x}_i - \mathbf{x}_j\|_2 = \ell_{ij}$ and calibrated spatial tolerance bubbles ($r_{\text{tol}} \approx 5\text{--}10\text{ cm}$), guaranteeing anatomical validity while synthesizing biological fluidity and facial co-articulation.
- **Backward Pass & Foundation Bridge (Video → Text):** Multi-channel perception (HaMeR 15-DOF hands, gaze) streams into an $\mathcal{O}(1)$ Causal Spatial Memory Bank (M_t) resolving regular, backward (INVITE), and orientation-only verbs (LOOK-AT) via aiming vectors $\vec{v}_{\text{aim}}$. An autoregressive foundation bridge (*Spatial-SignLLM*) unifies text, phonemes, and 3D coordinate tokens under causal language modeling loss $\mathcal{L}_{\text{LM}}$.

Scientific Alignment & Candidate Role Fit: The core challenge of sign language AI lies at the intersection of articulated kinematics and generative modeling: unconstrained neural models either hallucinate anatomically invalid poses (“rubber fingers”) or collapse spatial grammar. My background in continuous trajectory control, stochastic reinforcement learning (M.Sc. thesis, *bithax*), and 3D graphics kinematics directly prepares me to implement the physical modeling and programmatic compilers of Track A and Track C. However, mastering the generative residual and spatial grounding layers requires Dr. Farshad's specialized leadership in **topologically and geometrically constrained generative models** (*SCOPE*, *HieraSurg*) and **semantic scene graphs** (*SceneGenie*). Adapting Riemannian manifold projection ($\Pi_{\mathcal{M}}$) and graph topology to articulated skeletal trees provides the exact mathematical framework needed to ensure continuous diffusion preserves anatomical invariants while synthesizing fluid human micro-dynamics. This pairing creates an ideal supervisor-student synergy: my computational control and kinematic foundation complements your group's geometric deep learning expertise to develop clinically and socially impactful assistive digital twins.

4 4-Year Doctoral Execution Plan & Milestone Roadmap

To guarantee timely completion adhering to Aalto’s cumulative dissertation model (requiring 3 peer-reviewed first-author publications; *Aalto University Degree Regulations for Doctoral Degrees, SCI Guidelines on Article-Based Dissertations*), the doctoral roadmap is organized into two concurrent foundational pathways in Years 1–2 that converge into the central **Spatial-SignLLM (LSLM)** foundation model in Year 3, followed by cross-lingual transfer and community validation in Year 4:



High-Performance Computing: Pretraining on Motion-X++ (19.5M frames) and fine-tuning *Spatial-SignLLM* will natively leverage CSC’s EuroHPC LUMI supercomputer in Kajaani.

Linguistic Rights, WHO Well-Being & Participatory Evaluation: This doctoral research is fundamentally grounded in Finland’s Sign Language Act (*Viittomakielilaki 352/2015*), a world-leading statute that constitutionally recognizes sign language users and mandates public authorities to promote access to services in their mother tongue. We frame sign language accessibility strictly through the World Health Organization’s definition of health as “*a state of complete physical, mental, and social well-being and not merely the absence of disease.*” Linguistic exclusion creates severe communicative isolation that directly impairs mental and social health. Adhering to the principle of “*Nothing About Us Without Us,*” the architecture will undergo rigorous **empirical evaluations with native Deaf sign language users** in partnership with the Finnish Association of the Deaf (*Kuurojen Liitto ry*) as soon as possible—evaluating open-domain conversational fluency, avatar naturalism, and linguistic trust across everyday public and healthcare interactions.

5 Future Direction: Postdoctoral Trajectory & Societal Impact

Addressing the ELLIS selection criteria for long-term research vision, my postdoctoral objectives advance into two frontiers:

- ▶ **Embodied Physical Robotics & Non-Verbal HRI:** The core representations developed in this dissertation—sub-lexical phonological tokenization, closed-form 2-bone arm IK with humeral swivel orbits, and causal spatial memory—are directly portable to physical robots. As a postdoctoral researcher, I intend to investigate *Embodied Non-Verbal HRI*, transitioning from avatars to dual-arm humanoid robots (e.g., Boston Dynamics Atlas, Unitree, Franka Emika) for clinical rehabilitation, eldercare, and inclusive human-robot collaboration.
- ▶ **Open-Source European Foundation Model & Assistive Spin-off:** I aim to release an open-source European foundation model for low-resource sign languages (SVK, Finland-Swedish, LIS). Because our architecture executes at $< 100\text{ ms}$ streaming latency and $< 1\text{ KB/s}$ mesh bandwidth, I intend to establish an assistive venture deploying real-time sign translation onto edge mobile devices and spatial AR headsets (Apple Vision Pro, Meta Quest).

For complete mathematical derivations, proof of convergence, and kinematic benchmarks, please refer to our companion **Technical Research Foundation Proposal** (github.com/Joeytribb/sign-language-kinematics).

Thank you for your time and consideration. I welcome the opportunity to discuss how this research aligns with your group at Aalto and ELLIS Finland.

Sincerely,

Iniyan Andrews Joseph

Applicant, Doctoral Programme in Science (Doctor of Science in Technology)

Department of Computer Science, School of Science, Aalto University

Affiliated Research Unit: European Laboratory for Learning and Intelligent Systems (ELLIS) Institute Finland